Honeywell

Asset Performance Management

Upgrade Guide

Release 540.1

August 2023

DISCLAIMER

Contents

CHAPTER

1 ABOUT THIS GUIDE

1.1 Purpose

This guide describes the procedures for upgrading Asset Performance Management on the necessary server and client computers.

1.2 Audience

This document is intended for the project engineers and system administrators who are responsible for upgrading Asset Performance Management.

1.3 Revision history

Revision	Date	Description
1	August 2023	Asset Performance Management R540.1 General Availability (GA) release

1.4 References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

> **NOTE**
> Asset Performance Management documentation is available on the installation media in the following location:
>
> <Install Media>:\Installer\Components\US\Documents

Document name	Document number
Asset Performance Management Installation Guide	APMDOC-X389-en-R540.1
Asset Performance Management Configuration Guide	APMDOC-X391-en-R540.1
Asset Performance Management User's Guide	APMDOC-X392-en-R540.1

For the most up-to-date version of the documentation, see **https:// www.honeywellprocess.com** (click the Support tab). User registration and login are required to access the Honeywell Process site.

CHAPTER

2 INTRODUCTION

This guide provides instructions for upgrading Asset Performance Management to version R531.1, whether upgrading is in the same servers or migrating your system to a new server.

> **NOTE**
> Migrating to a new system requires a fresh installation of Asset Performance Management on the new server. For detailed instructions for installing Asset Performance Management, refer the Asset Performance Management **Installation Guide**.

2.1 Supported Upgrade Versions

2.1.1 In-place Upgrade Supported Versions

In-place upgrade is supported for systems using R520.x, R530.x, or R531.x to R532.1 on windows server 2016/2019. Other existing Asset Performance Management systems can be migrated to new Windows Server 2016/2019 systems.

> **NOTE**
> Any versions prior to R511.X should be migrated to R511.X by running respective database migration tool and then upgrade to R532.1. For upgrade instructions to 531.1, refer to the R531.1 installation.

See the below sections for instructions on upgrading the existing system:

- In-place Upgrade Split Servers
- In-place Upgrade Primary and Shadow Servers
- In-place Upgrade Single Server
- Upgrading Client Tools

2.1.2 System Migration or Database Restore

Migrating to a new system involves restoring the database on a new server. Database restoration is supported from Asset Manager R410.2 or Asset Performance Management R520.x, R530.x, or R531.x to R532.1.

Refer to the following sections for instructions on migrating to a new system or restoring to a database:

- Migrating to a New System.
- Restoring a Database.

CHAPTER

3 In-place Upgrade Split Servers

3.1 Prepare a Split Server for Upgrade

This chapter covers the preparations to be done on all servers in a split server topology before upgrading the servers.

> NOTE
> The protocol used on the servers (http or https) cannot be changed during migration. If the existing servers use http, http should be selected and if the existing servers use https, https should be selected during the upgrade.

3.1.1 Confirm User and Services Settings

The following are the user and service settings requirements:

1. You are a Local Administrator and have SQL Server (SysAdmin) privileges.
2. Net 4.8 is installed on the system.
3. The Windows Update service is disabled. (**Run > Services**, right-click **Windows Update** and then select **Disable**.)
4. Ensure you have a Services User that does not require login permissions to the Asset Performance Management system.

   > ATTENTION
   > Do not use the installation (current) user as the Services User. If the selected Services User has login permissions, they will be removed as a part of the Asset Performance Management installation.
   >
   > Once the installation is complete, the Services User will no longer be able to login to the system locally or remotely.

5. Ensure the services (runtime) are added to the MESDBUsers Windows Group.
6. Ensure all required users are added to the Windows Reliability Engineers group.
7. Navigate to **<Install Location>:\ProgramData\Microsoft\Crypto\RSA\MachineKeys** and ensure the services account is given the following permissions to the folder on the Primary Server:
   - Read & Execute
   - List Folder contents
   - Read

3.1.2 Confirm SQL Server Permissions and Settings

- Ensure the installation user has db_owner permission to the APM AssetTaskDataModel database.
- Ensure the installation user has public, db_datareader and db_datawriter on all Honeywell MES databases.
- Ensure the MESDBUser has public, db_datareader and db_datawriter permissions to the APM AssetTaskDataModel database.
- Delete the services user account in SQL.
- The migration adds the historical data clean-up SQL job (see "Configure Historized Data Clean-up" in the Asset Performance Management Configuration Guide for details). Ensure the SQL Server Agent service is running either during or after the migration for this change to be completed.

> **ATTENTION**
> Ensure SQL replication is disabled before beginning the migration.

3.1.3 Install all Microsoft Updates

1. Run Microsoft Updates to ensure all current updates are installed.
2. Restart the server.

3.1.4 Stop Services

Stop the following services in Application, Web and Calculation service

- AMApplicationServer (.NET Service for AM Application Server)
- AssetManagerIntegrationService (.NET Service for AM Integration Service)
- Honeywell MES Event Processor (Honeywell Intuition Event Processing Service)
- Honeywell UAS Calculation Host
- Honeywell UAS Event Translator Service
- Honeywell Forge Bulk Model Importer Service
- Honeywell Forge Bulk Model Exporter Service
- All Uniformance PHD services
- Close all the Asset Performance Management related applications, along with the browser window.
- If Uniformance Insight application is already installed then you need to stop the below service:

  Honeywell Content Indexing Service

> **NOTE**
> Make sure no APM or Uniformance PHD applications are running in Task Manager

> **NOTE**
> Change the Start Mode of NotificationConfig4.0 IIS application pool to *OnDemand * from *Running * before you start Migration.

3.1.5 Perform IIS reset

Perform IIS reset in the following nodes

- App server
- web server
- calc
- single server

NotificationConfig4.0 APP Pool start mode is to be changed from “alwaysRunning” to “OnDemand”

1. Go to Run and type “INETMGR”.
2. Click **Ok** and go to Application Pools.
3. Right click “**NotificationConfig4.0**” and select Advanced Settings.
4. Change Start mode from “alwaysRunning” to “OnDemand” and click **Ok**.
5. Perform **IISReset**.

3.1.6 Back-up Existing Files

The upgrade copies your database information and files automatically. However, creating a back-up copy of the database and any custom files is recommended before beginning the upgrade.

> NOTE
> For instructions on backing-up a database through SQL Server Management Studio, see the SQL Server Management Studio user documentation.

1. Take back up from

   [InstallDrive]:\Program File (x86)\Honeywell\MES\AssetManager\AppServer\Bin\Honeywell.AM.Service.AMHostingService.exe.config

   [InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\CalcHost\Honeywell.UAS.CalculationWorker.exe.config
2. Browse to **[InstallDrive]:\Honeywell\MES\Core\WWWMESCore\ AMInformationFolders\,** copy the folders: **Images, Icons, and Documents**.
3. Paste the copy of the folders to any folder or external drive on your computer.

3.1.7 Install Microsoft SQL Server 2012 Native Client

1. On the installation media, browse to installer/prerequisites/sqlncli.msi
2. Right click sqlncli.msi and click **install**.
3. Follow on screen instructions to complete the installation.

3.2 Upgrade a Database Server

This chapter describes how to upgrade a Database Sever in a split server topology.

> ATTENTION
> In-place migration is only supported on Windows Server 2016/2019 systems. If your system is installed on Windows Server 2012 R2, see "Upgrading on a Windows Server R2 Server" in the Asset Performance Management Troubleshooting Guide before continuing.

> **NOTE**
> During the installation TLS 1.2 is enabled on the server.

> **NOTE**
> Due to security changes, during the upgrade new security certificates must be applied. You will be prompted to configure the certificates during the upgrade process.

3.2.1 Upgrading on an Existing Database Server

To grade a Asset Performance Management Database Server

1. Perform an iisreset.
2. On the installation media of the new data base server, right-click **setup.exe**, and then select **Run as Administrator**.
3. In the RabbitMQ prompt, click **No**.
4. On the Welcome page, click **Next**.
5. Read through the license agreement and click **I accept the terms in the license agreement** if you agree to the license terms and then click **Next**.
6. The components for the server you are installing are selected and cannot be cleared. Click **Next**.
7. On the **Features and Options Selection** page, the features required for your components are already selected, click **Next**.
8. Do all of the following:
    a. In the **Login Information for Application Services** area, in the **Account Password** box, type the password for the Asset Performance Management services user.
    b. (Optional) If you are using an existing EPHD installation, in the **EPHD Server Details** area, in the **EPHD Server Name** box, type the name of the EPHD server.

    > **NOTE**
    > EPHD does not support SQL Clustering and non-default collation.

    c. Enter the Domain User Group Name. The domain user group name should be in the doamin\UserGroupName format. For example: l4-dc\UASServiceGroup.
    d. In the Service Account Permissions area, select **I accept the service account must be denied the ability to logon to the desktop to continue.**

    > **NOTE**
    > The installation cannot continue if the service account has login permissions to the desktop. This check box must be selected for the installation to succeed.
    >
    > Once this change is made, the service account can no longer log on to the computer (locally or remotely) and cannot be used to run programs that interact with the desktop, such as or Internet Explorer.

    e. Click **Next**.

9. Click **Install** to start the installation.
10. Click **Finish**.

3.3 Upgrade an Application Server

This chapter how to upgrade an Application Sever in a split server topology.

> **ATTENTION**
> In-place migration is only supported on Windows Server 2016 systems. If your system is installed on Windows Server 2012 R2, see "Upgrading on a Windows Server R2 Server" in the Asset Performance Management Troubleshooting Guide before continuing.

> **NOTE**
> During the installation TLS 1.2 is enabled on the server.

> **NOTE**
> Due to security changes, during the upgrade new security certificates must be applied. You will be prompted to configure the certificates during the upgrade process.

3.3.1 Upgrading on an Existing Application Server

To upgrade a Asset Performance Management Application Server

1. Perform an iisreset.
2. On the installation media of the new data base server, right-click **setup.exe**, and then select **Run as Administrator**.
3. In the RabbitMQ prompt, click **No**.
4. On the Welcome page, click **Next**.
5. Read through the license agreement and click **I accept the terms in the license agreement** if you agree to the license terms and then click **Next**.
6. The components for the server you are installing are selected and cannot be cleared. Click **Next.**
7. On the **Features and Options Selection** page, the features required for your components are already selected, click **Next**.
8. In the **Intuition Environment Setup**, correct any issues, and then click **Exit**. (If the Exit button does not appear with the success message, close the browser window to continue.)

   For example, if Certificate Configuration is flagged, do all of the following:

    a. Beside **Certificate Configuration**, click **Repair**.
    b. Select .
    c. Click **Configure**.
    d. In the **Success** dialog box, click **OK**.
    e. Click the **Home** tab and click **Repair all**.

Do all of the following:

a. In the **Login Information for Application Services** area, in the **Account Password** box, type the password for the Asset Performance Management services user.

b. Enter the Domain User Group Name. The domain user group name should be in the domain\UserGroupName format. For example: I4-dc\UASServiceGroup.

c. In the Service Account Permissions area, select **I accept the service account must be denied the ability to logon to the desktop to continue**.

> **NOTE**
> The installation cannot continue if the service account has logon permissions to the desktop. This check box must be selected for the installation to succeed. Once this change is made, the service account can no longer log on to the computer (locally or remotely) and cannot be used to run programs that interact with the desktop, such as or Internet Explorer

d. Click **Next**.

9. Do all of the following:

   a. In the **Website selection** box, ensure the web site you are using for Asset Performance Management is selected.

   b. In the **Calc Server Name** box, type the name of the main Calculation Server.

   c. Click **Next**.

10. Do all of the following:

    a. In the **Web Server Host Name** box, type the name of the Web Server.

    > **ATTENTION**
    > If you are using https, enter the fully qualified domain name of the Web Server.

    b. In the **Web Server HTTP Port No** box, type the port number used on the Web Server.

    c. In the **Web Server Protocol Name** box, select the protocol (**http** or **https**) used on the Web Server.

    d. Click **Next**.

11. On the Email Dispatcher page, do all of the following:

    a. In the **Sender's Email ID**, type the email address that will appear in the From field of emails sent by the system.

    b. In the **Default Email ID**, type the email address that will receive a notification if an email delivery fails.

    c. In the SMTP Server Name box, type the name of the SMTP server.

    d. You can also enable SSL, by clicking the Enable SSL check box and provide the following information

    - Server Name of SMTP server.
    - Port Number of the SMTP server.
    - Secure Socket Option - Enter socket option. For more details refer Configuring Secure Socket Option from Asset Performance Management Installation Guide.

    e. Click **Next**.

12. Click **Install** to start the installation.

> **NOTE**
> When running the upgrade on the Application Server, a notice that PkgMgr.exe is deprecated can appear during the installation. If this message appears, click OK to continue the migration.

13. Click **Finish**.

3.4 Upgrade a Web Server

This chapter how to upgrade a Web Sever in a split server topology.

> **ATTENTION**
> In-place migration is only supported on Windows Server 2016 systems. If your system is installed on Windows Server 2012 R2, see "Upgrading on a Windows Server R2 Server" in the Asset Performance Management Troubleshooting Guide before continuing.

> **NOTE**
> During the installation TLS 1.2 is enabled on the server.

> **NOTE**
> Due to security changes, during the upgrade new security certificates must be applied. You will be prompted to configure the certificates during the upgrade process.

3.4.1 Upgrading on an Existing Web Server

To update a Asset Performance Management Web Server

1. Perform an iisreset.
2. On the installation media of the new web server, right-click setup.exe, and then select Run as Administrator.
3. In the RabbitMQ prompt, click No.
4. On the Welcome page, click Next.
5. Read through the license agreement and click I accept the terms in the license agreement if you agree to the license terms and then click Next.
6. The components for the server you are installing are selected and cannot be cleared. Click Next.
7. On the Features and Options Selection page, the features required for your components are already selected, click Next.
8. In the Intuition Environment Setup, correct any issues, and then click Exit. (If the Exit button does not appear with the success message, close the browser window to continue.)

   For example, if Certificate Configuration is flagged, do all of the following:

   a. Beside **Certificate Configuration**, click **Repair**.
   b. Select Web Components.
   c. Click Configure.
   d. In the Success dialog box, click OK.
   e. Click the Home tab and click Repair all.

On the Notification and DAS Configuration page, do all of the following:

f. Under the CMMS Alert section, the default path where the CMMS XML files are saved is displayed. If you want to change the default path, perform the following steps:
    a. Click Browse.
    b. The Browse for Folder page appears.
    c. Select the path where you want save the CMMS XML files.
    d. Click OK.

g. Click Next.

9. Do all of the following:
    a. In the Login Information for Application Services area, in the Account Password box, type the password for the Asset Performance Management services user.
    b. Enter the Domain User Group Name. The domain user group name should be in the domain\UserGroupName format. For example: l4-dc\UASServiceGroup.
    c. In the Service Account Permissions area, select I accept the service account must be denied the ability to logon to the desktop to continue.

    > **NOTE**
    > The installation cannot continue if the service account has logon permissions to the desktop. This check box must be selected for the installation to succeed. Once this change is made, the service account can no longer log on to the computer (locally or remotely) and cannot be used to run programs that interact with the desktop, such as or Internet Explorer.

    d. Click Next.

10. Do all of the following:
    a. In the Website selection box, ensure the web site you are using for Asset Performance Management is selected.
    b. In the Calc Server Name box, type the name of the main Calculation Server.
    c. Click Next.

11. Do all of the following:
    a. In the Application Server Host Name, type the name of the Application Server.
    b. In the Application Server Port Number, type the port number used on the Application Server.
    c. In the **Application Server Protocol Name**, select the protocol (http or https) used on the Application Server.
    d. In the **Application Server NET.TCP Port Number**, type the net.tcp port number used on the Application Server.
    e. Click **Next**.

12. On the Email Dispatcher page, do all of the following:
    a. In the **Sender’sEmail ID**, type the email address that will appear in the From field of emails sent by the system.
    b. In the **Default Email ID**, type the email address that will receive a notification if an email delivery fails.
    c. In the SMTP Server Name box, type the name of the SMTP server.

d. You can also enable SSL, by clicking the Enable SSL check box and provide the following information:

   a. Server Name of SMTP server.

   b. Port Number of the SMTP server.

   c. Secure Socket Option - Enter socket option. For more details refer Configuring Secure Socket Option from Asset Performance Management Installation Guide.

e. Click **Next**.

13. Click **Install** to start the installation.
14. Click **Finish**.

3.5 Upgrade a Calculations Server

> **ATTENTION**
> In-place migration is only supported on Windows Server 2016/2019 systems. If your system is installed on Windows Server 2012 R2, see "Upgrading on a Windows Server R2 Server" in the APM Troubleshooting Guide before continuing.

> **NOTE**
> During the installation TLS 1.2 is enabled on the server.

> **NOTE**
> Due to security changes, during the upgrade new security certificates must be applied. You will be prompted to configure the certificates during the upgrade process.

3.5.1 Upgrading on an Existing Calculation Server

To upgrade a Asset Performance Management Calculation Server

1. Perform an iisreset.
2. On the installation media of the new data base server, right-click **setup.exe**, and then select **Run as Administrator**.
3. If RabbitMQ is not already installed, a prompt appears. In the RabbitMQ prompt, click **Yes**.If RabbitMQ needs to be installed, when it completes, reopen the installation wizard as described in step 2.
4. On the Welcome page, click **Next**.
5. Read through the license agreement and click **I accept the terms in the license agreement** if you agree to the license terms and then click **Next**.
6. The components for the server you are installing are selected and cannot be cleared. Click **Next**.
7. On the **Features and Options Selection** page, the features required for your components are already selected, click **Next**.
8. In the **Intuition Environment Setup,** correct any issues, and then click **Exit**. (If the Exit button does not appear with the success message, close the browser window to continue.)

   For example, if Certificate Configuration is flagged, do all of the following:

   a. Beside **Certificate Configuration**, click **Repair**.
   b. Select.
   c. Click **Configure**.
   d. In the **Success** dialog box, click **OK**.
   e. Click the **Home** tab and click **Repair all**.

9. On the **Notification and DAS Configuration** page, do all of the following:
   a. Under the **CMMS Alert** section, the default path where the CMMS XML files are saved is displayed. If you want to change the default path, perform the following steps:
      a. Click **Browse**.
      b. The **Browse for Folder** page appears.
      c. Select the path where you want save the CMMS XML files.
      d. Click **OK**.

   b. Click **Next**.

10. Do all of the following:
   a. In the **Login Information for Application Services** area, in the **Account Password** box, type the password for the Asset Performance Management services user.
   b. If you are using an existing EPHD installation, in the **EPHD Server Details** area, in the **EPHD Server Name** box, type the name of the EPHD server.

   > NOTE
   > EPHD does not support SQL Clustering.

   c. Enter the Domain User Group Name. The domain user group name should be in the doamin\UserGroupName format. For example: l4-dc\UASServiceGroup.
   d. In the Service Account Permissions area, select **I accept the service account must bedenied the ability to logon to the desktop to continue**.

   > NOTE
   > The installation cannot continue if the service account has logon permissions to the desktop. This check box must be selected for the installation to succeed. Once this change is made, the service account can no longer log on to the computer (locally or remotely) and cannot be used to run programs that interact with the desktop, such as or Internet Explorer.

   e. Click **Next**.

11. Do all of the following:
   a. In the **Website selection** box, ensure the web site you are using for Asset Performance Management is selected.
   b. Click **Next**.

12. On the Email Dispatcher page, do all of the following:

a. In the **Sender'sEmail ID**, type the email address that will appear in the From field of emails sent by the system.
b. In the **Default Email ID**, type the email address that will receive a notification if an email delivery fails.
c. In the SMTP Server Name box, type the name of the SMTP server.
d. You can also enable SSL, by clicking the Enable SSL check box and provide the following information
    a. Server Name of SMTP server.
    b. Port Number of the SMTP server.
    c. Secure Socket Option - Enter socket option. For more details refer Configuring Secure Socket Option from Asset Performance Management Installation Guide.
e. Click **Next**.

13. Click **Install** to start the installation.
14. Click **Finish**.

3.5.2 Post Installation Steps

On the additional calculation server, restart the services in the following order:

- AMApplicationServer
- Honeywell UAS Calculation Host

The following services can then be stopped on the additional calculation server if they are running:

- RabbitMQ
- Honeywell UAS Event Translator Service

3.6 Finalize a Split Server Upgrade

This chapter describes the steps to take after completing the upgrade on servers in a split server topology

3.6.1 Confirm User Permissions and Settings

Confirm the follow are still true, and add the required permissions if not:

- Ensure the MESDBUser has Public, DBreader and DBwriter permissions to the APM AssetTaskDataModel database.
- Ensure the installation user has db owner permission to the APM AssetTaskDataModel database.
- Ensure all required users are added to the Windows Reliability Engineers group.
- Ensure Directory Browsing is enabled on the web site in IIS.

> **ATTENTION**
> After the upgrade to R532.1, users must clear their browser cache for some screens (including configuration) to load as expected.

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3.6.2 Update the Database Index

1. Open SQL Server Management Studio: **Start > All Programs > Microsoft SQL Server [Version Number] > SQL Server Management Studio**

   The Microsoft SQL Server Management Studio window appears.
2. Enter the details for the database instance to be used by Asset Performance Management, and then click **Connect**.
3. Run the following SQL script:

   [InstallDrive]:\Installer\Components\UAS\Installers\AssetCX\Scripts\BulkConfig_All_ViewIndexs.sql

CHAPTER

4 IN-PLACE UPGRADE SINGLE SERVER

4.1 Prepare a Single Server for Upgrade

This chapter describes how to upgrade Asset Performance Management R520.1 to Asset Performance Management R530.1.

4.1.1 Confirm User and Services Settings

- The installation user is a Local Administrator and has SQL Server (SysAdmin) privileges.
- Net 4.7.2 is installed on the system.
- The Windows Update service is disabled. (**Run** > **Services**, right-click **Windows Update** and then select **Disable**.)
- Ensure you have a Services User that does not require logon permissions to the Asset Performance Management computer.

> **ATTENTION**
> Do not use the installation (current) user as the Services User. If the selected Services User has logon permissions, they will be removed as a part of the Asset Performance Management installation. Once the installation is complete, the Services User will no longer be able to log on to the computer locally or remotely.

- Ensure the services (runtime) is added to the MESDBUsers Windows Group.
- Ensure all required users are added to the Windows Reliability Engineers group (Add required users to the Windows Reliability Engineers group).
- Ensure the services account is given the following permissions to the [InstallDrive]:\ProgramData\Microsoft\Crypto\RSA\MachineKeys folder on the Primary Server:
    - Read & Execute
    - List Folder contents
    - Read

4.1.2 Confirm SQL Server Permissions and Settings

- Ensure the installation user has db_owner permission to the APM AssetTaskDataModel database.
- Ensure the installation user has public, db_datareader and db_datawriter on all Honeywell MES databases.
- Ensure the MESDBUser has public, db_datareader and db_datawriter permissions to the APMAssetTaskDataModel database.
- Delete the services user account in SQL.
- The migration adds the historical data clean-up SQL job (see “Configure Historized Data Clean-up” in the Asset Performance Management Configuration Guide for details). Ensure the SQL Server Agent service is running either

during or after the migration for this change to be completed.

> **ATTENTION**
> Ensure SQL replication is disabled before beginning the migration.

4.1.3 Install all Microsoft Updates

1. Run Microsoft Updates to ensure all current updates are installed.
2. Restart the server.

4.1.4 Stop Services

Stop the following services:

- AMApplicationServer (.NET Service for AM Application Server)
- AssetManagerIntegrationService (.NET Service for AM Integration Service)
- Honeywell MES Event Processor (Honeywell Intuition Event Processing Service)
- Honeywell UAS Calculation Host
- Honeywell UAS Event Translator Service
- Honeywell Forge Bulk Model Importer Service
- Honeywell Forge Bulk Model Exporter Service
- All Uniformance PHD services
- Close all the Asset Performance Management related applications, along with the browser window.
- If Uniformance Insight application is already installed then you need to stop the below service:
  Honeywell Content Indexing Service

> **NOTE**
> Make sure no APM or Uniformance PHD applications are running in Task Manager

4.1.5 Perform IIS reset

Perform IIS reset in the following nodes

- App server
- web server
- calc
- single server

NotificationConfig4.0 APP Pool start mode is to be changed from “alwaysRunning” to “OnDemand”

1. Go to Run and type “INETMGR”.
2. Click **Ok** and go to Application Pools.
3. Right click “**NotificationConfig4.0**” and select Advanced Settings.
4. Change Start mode from “alwaysRunning” to “OnDemand” and click **Ok**.
5. Perform **IISReset**.

4.1.6 Back-up Existing Files

The upgrade copies your database information and files automatically. However, creating a back-up copy of the database and any custom files is recommended before beginning the upgrade.

> **NOTE**
> For instructions on backing-up a database through SQL Server Management Studio, see the SQL Server Management Studio user documentation.

1. Take back up from

   [InstallDrive]:\Program File (x86)\Honeywell\MES\AssetManager\AppServer\Bin\Honeywell.AM.Service.AMHostingService.exe.config

   [InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\CalcHost\Honeywell.UAS.CalculationWorker.exe.config
2. Browse to **[InstallDrive]:\Honeywell\MES\Core\WWWMESCore\AMInformationFolders**, copy the folders: **Images**, **Icons**, and **Documents**.
3. Paste the copy of the folders to any folder or external drive on your computer.

4.1.7 Install Microsoft SQL Server 2012 Native Client

1. On the installation media, browse to installer/prerequisites/sqlncli.msi
2. Right click sqlncli.msi and click **install**.
3. Follow on screen instructions to complete the installation.

4.2 Upgrading a Single Server

This chapter covers how to upgrade a Single Server.

> **ATTENTION**
> After the upgrade to R530.1, users must clear their browser cache for some screens (including configuration) to load as expected.

> **ATTENTION**
> In-place migration is only supported on Windows Server 2016 systems. If your system is installed on Windows Server 2012 R2, see “Upgrading on a Windows Server R2 Server” in the APM Troubleshooting Guide before continuing.

> **NOTE**
> For instructions on backing-up a database through SQL Server Management Studio, see the SQL Server Management Studio user documentation.

4.2.1 Upgrading an existing Single Server

> **NOTE**
> The protocol used on the server (http or https) cannot be changed during migration. If the existing server uses http, http must be selected during the upgrade. If the existing server uses https, https must be selected during the upgrade.

> **ATTENTION**
> In-place migration is only supported on Windows Server 2016 systems.

To upgrade a Asset Performance Management Single Server

1. Perform an iisreset.
2. On the installation media of the new data base server, right-click **setup.exe**, and then select **Run as Administrator**.
3. If RabbitMQ is not already installed, a prompt appears. In the RabbitMQ prompt, click **Yes**.

   If RabbitMQ needs to be installed, when it completes, reopen the installation wizard as described in step 2.
4. On the Welcome page, click **Next**.
5. Read through the license agreement and click **I accept the terms in the licenseagreement**if you agree to the license terms and then click **Next**.
6. On the Migration Information page, click **Next**.
7. The components for the server you are installing are selected and cannot be cleared. Click Next.
8. On the **Features and Options Selection** page, the features required for your components are already selected, click **Next**.
9. In the **Intuition Environment Setup,** correct any issues, and then click **Exit**. (If the Exit button does not appear with the success message, close the browser window to continue.)

   For example, if Certificate Configuration is flagged, do all of the following:
   a. Beside **Certificate Configuration**, click **Repair**.
   b. Select **Web Components** and enter the Application server name.
   c. Click **Configure**.
   d. In the **Success** dialog box, click **OK**.
   e. Click the **Home** tab and click **Repair all**.
10. On the **Notification and DAS Configuration** page, do all of the following:
    a. Under the **CMMS Alert** section, the default path where the CMMS XML files are saved is displayed. If you want to change the default path, perform the following steps:
       a. Click **Browse**.

          The **Browse for Folder** page appears.
       b. Select the path where you want save the CMMS XML files.
       c. Click **OK**.
    b. Click **Next**.
11. Do all of the following:

a. In the **Login Information for Application Services** area, in the **Account Password** box, type the password for the Asset Performance Management services user.

b. (Optional) If you are using an existing EPHD installation, in the **EPHD Server Details** area, in the **EPHD Server Name** box, type the name of the EPHD server.

> **NOTE**
> EPHD does not support SQL Clustering.

c. In the SQL Cluster Support area, select Enable SQL Cluster to enable SQL Cluster deployment.

d. Enter the Domain User Group Name. The domain user group name should be in the doamin\UserGroupName format. For example: I4-dc\UASServiceGroup.

e. In the Service Account Permissions area, select **I accept the service account must be denied the ability to logon to the desktop to continue.**

> **NOTE**
> The installation cannot continue if the service account has logon permissions to the desktop. This check box must be selected for the installation to succeed.
>
> Once this change is made, the service account can no longer log on to the computer (locally or remotely) and cannot be used to run programs that interact with the desktop, such as or Internet Explorer.

f. Click **Next**.

12. Do all of the following:
    a. In the **Website selection** box, ensure the web site you are using for Asset Performance Management is selected.
    b. In the **Calc Server Name** box, type the name of the main Calculation Server.
    c. Click **Next**.

13. On the Email Dispatcher page, do all of the following:
    a. In the **Sender'sEmail ID**, type the email address that will appear in the From field of emails sent by the system.
    b. In the **Default Email ID**, type the email address that will receive a notification if an email delivery fails.
    c. In the **SMTP Server Name** box, type the name of the SMTP server.
    d. In the **SMTP Port Number**, type the port number used on the SMTP server.
    e. Click **Next**.

14. Click **Install** to start the installation.
15. Click **Finish**.
16. If the system has other Honeywell products in stalled (such as Uniformance Insight), run the **SystemCatalogSeedUtility.exe** file located in the following folder:

```
[install drive]:\Program Files
(x86)\Honeywell\MES\Tools\CAM\SystemCatalogSeedUtility\
```

17. Do all of the following to update the calculations:

a. Ensure the AMApplicationServer service is running.

b. With a user that belongs to the Reliability Engineers group (see “Add required users to the Windows Reliability Engineers group” on page 154), run the **SentinelUpgradeConsoleUtil.exe** file, located in the following folder:

```
[Install Media]\Utilities\Sentinel_Upgrade_Utility\
```

See Model Migration Utility Parameters for the parameters used to run the migration utility.

Model Migration Utility Parameters

Use the following syntax to run the model migration utility:

```
SentinelUpgradeConsoleUtil --address [ServerName] --[protocol] --port [PortNumber]
```

Examples:

```
SentinelUpgradeConsoleUtil --address localhost --port 80
SentinelUpgradeConsoleUtil --address webserver1.oursite.com --https --port 8000
```

> **ATTENTION**
> If the protocol is https, localhost cannot be used as the server address.

To see a list of parameters while using the model migration utility, use --help to run the utility:

```
SentinelUpgradeConsoleUtil --help
```

4.3 Finalize a Single Server Upgrade

This chapter describes how to upgrade Asset Performance Management R520.1 to Asset Performance Management R540.1.

4.3.1 Confirm User Permissions and Settings

Confirm the follow are still true, and add the required permissions if not:

- Ensure the MESDBUser has Public, DBreader and DBwriter permissions to the APM AssetTaskDataModel database.
- Ensure the installation user has db owner permission to the APM AssetTaskDataModel database.
- Ensure all required users are added to the Windows Reliability Engineers group (see Add required users to the Windows Reliability Engineers group).
- Ensure Directory Browsing is enabled on the web site in IIS (see Ensure Directory Browsing is enabled for the site in IIS).

> **ATTENTION**
> After the upgrade to R540.1, users must clear their browser cache for some screens (including configuration) to load as expected.

4.3.2 Update the Database Index

1. Open SQL Server Management Studio: **Start > All Programs > Microsoft SQL Server [Version Number] > SQL Server Management Studio**

   The Microsoft SQL Server Management Studio window appears.

2. Enter the details for the database instance to be used by Asset Performance Management, and then click **Connect**.
3. Run the following SQL script:

   [InstallDrive]:\Installer\Components\UAS\Installers\AssetCX\Scripts\BulkConfig_All_ViewIndexs.sql

4.3.3 Add required users to the Windows Reliability Engineers group

1. **Start > Server Manager**.
2. On the server tree, choose **Configuration** > **Local Users and Groups** > **Groups**.
3. Right-click the **Reliability Engineers** group, and click **Add to Group**.
4. In the **Reliability Engineers Properties** dialog box, click **Add**.
5. Enter all the names of all of the following users:
   a. The services (runtime) user.

      > **ATTENTION**
      > If the services user is not added to the Reliability Engineers group, some features will not function as expected.

   b. The install user.
   c. All the users who will need to use the APM Configuration Excel Plugin and the Asset Performance Management Configuration.
6. Click **OK**.

   The user names appear in the group list.

7. Click **OK** to close the Reliability Engineers Properties dialog box.

4.3.4 Ensure Directory Browsing is enabled for the site in IIS

1. **Start > All Apps > Internet Information Services (IIS) Manager**

   Internet Information Services (IIS) Manager appears.

2. Expand Sites, and select the web site used for Asset Performance Management .
3. On Default Web Site Home page, double-click Directory Browsing.
4. In the right-hand side bar, ensure Directory Browsing is enabled.

   If it is disabled an alert reading “Directory browsing has been disabled” is visible and the Enable button is available under Actions.

5. If it is disabled, click **Enable**.
6. Perform an iisreset.

4.3.5 Update the Internal Data Source Connection Strings

If you upgraded a database and ran the Migration Utility, you need to upgrade the connections strings for the internal data sources.

> **ATTENTION**
> If the connection strings are not updated, asset metrics data can be lost on a metrics reset.

1. Open **SQL Server Management Studio**.
2. Expand **Databases** >**Honeywell.MES.AssetTask.DataModel.AssetTaskDataModel**>**Tables**>**ExternalDataSources**.
3. Edit the table rows.
4. In the **Internal State Storage** row, in the ConnectionInformation column, replace (local) with the name of your Single Server.
5. In the **Internal Historical Storage**, in ConnectionInformation column, replace (local) with the name of your Single Server.

4.3.6 Systems Running EPHD

After installing Asset Performance Management and updating the EPHD settings, the EPHD system is left in an Initializing state. To bring the state back to active, run the following commands:

```
phdctl stop /y
phdctl stop /y dbsecurity
phdctl stop /y udbserver
phdctl start cold
```

4.3.7 Restoring EPHD archive data

To restore EPHD machine data,

1. Stop all the Uniformance PHD services.
2. Go to services, right click Uniformance PHD Server and slect properties.
3. Click General and navigate to the path from path to executable.
4. Click Archive and rename all the file name from CHAR00001 to CHAR00002.
5. Go to :\Program Files (x86)\Honeywell\Uniformance\PHDServer\Archive and copy all the files from the location.
6. Navigate back to Path to executable location of the Uniformance PHD Server properties and paste.
7. Start all the Uniformance PHD Services back.

CHAPTER

5 IN-PLACE UPGRADE PRIMARY AND SHADOW

5.1 Prepare a Primary or Shadow Server for Upgrade

This chapter explains how to upgrade Asset Performance Management R520.1 to Asset Performance Management R530.1.

5.1.1 Confirm User and Services Settings

- The installation user is a Local Administrator and has SQL Server (SysAdmin) privileges. o .
- Net 4.7.2 is installed on the system.
- The Windows Update service is disabled. (**Run** > **Services**, right-click **Windows Update** and then select **Disable**.)
- Ensure you have a Services User that does not require logon permissions to the Asset Performance Management computer.

> **ATTENTION**
> Do not use the installation (current) user as the Services User. If the selected Services User has logon permissions, they will be removed as a part of the Asset Performance Management installation. Once the installation is complete, the Services User will no longer be able to log on to the computer locally or remotely.

- Ensure the services (runtime) is added to the MESDBUsers Windows Group.
- Ensure all required users are added to the Windows Reliability Engineers group (Add required users to the Windows Reliability Engineers group).
- Ensure the services account is given the following permissions to the [InstallDrive]:\ProgramData\Microsoft\Crypto\RSA\MachineKeys folder on the Primary Server:
  - Read & Execute
  - List Folder contents
  - Read

5.1.2 Confirm SQL Server Permissions and Settings

- Ensure the installation user has db_owner permission to the APM AssetTaskDataModel database.
- Ensure the installation user has public, db_datareader and db_datawriter on all Honeywell MES databases.
- Ensure the MESDBUser has public, db_datareader and db_datawriter permissions to the APM AssetTaskDataModel database.
- Delete the services user account in SQL.
- The migration adds the historical data clean-up SQL job (see “Configure Historized Data Clean-up” in the Asset Performance Management Configuration Guide for details). Ensure the SQL Server Agent service is running either

during or after the migration for this change to be completed.

> **ATTENTION**
> Ensure SQL replication is disabled before beginning the migration.

5.1.3 Install all Microsoft Updates

1. Run Microsoft Updates to ensure all current updates are installed.
2. Restart the server.

5.1.4 Stop Services

Stop the following services:

- AMApplicationServer (.NET Service for AM Application Server)
- AssetManagerIntegrationService (.NET Service for AM Integration Service)
- Honeywell MES Event Processor (Honeywell Intuition Event Processing Service)
- Honeywell UAS Calculation Host
- Honeywell UAS Event Translator Service
- Honeywell Forge Bulk Model Importer Service
- Honeywell Forge Bulk Model Exporter Service
- All Uniformance PHD services
- Close all the Asset Performance Management related applications, along with the browser window.
- If Uniformance Insight application is already installed then you need to stop the below service: Honeywell Content Indexing Service

> **NOTE**
> Make sure no APM or Uniformance PHD applications are running in Task Manager.

5.1.5 Perform IIS reset

Perform IIS reset in the following nodes

- App server
- web server
- calc
- single server

NotificationConfig4.0 APP Pool start mode is to be changed from “alwaysRunning” to “OnDemand”

1. Go to Run and type “INETMGR”.
2. Click **Ok** and go to Application Pools.
3. Right click “**NotificationConfig4.0**” and select Advanced Settings.
4. Change Start mode from “alwaysRunning” to “OnDemand” and click **Ok**.
5. Perform **IISReset**.

5.1.6 Back-up Existing Files

The upgrade copies your database information and files automatically. However, creating a back-up copy of the database and any custom files is recommended before beginning the upgrade.

> NOTE
> For instructions on backing-up a database through SQL Server Management Studio, see the SQL Server Management Studio user documentation.

1. Take back up from

   [InstallDrive]:\Program File (x86)\Honeywell\MES\AssetManager\AppServer\Bin\Honeywell.AM.Service.AMHostingService.exe.config

   [InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\CalcHost\Honeywell.UAS.CalculationWorker.exe.config
2. Browse to **[InstallDrive]:\Honeywell\MES\Core\WWWMESCore\AMInformationFolders**, copy the folders: **Images**, **Icons**, and **Documents**.
3. Paste the copy of the folders to any folder or external drive on your computer.

5.2 Upgrading a Primary Server

This chapter covers how to upgrade Asset Performance Management R520.1 to Asset Performance Management R530.1.

> ATTENTION
> After the upgrade to R530.1, users must clear their browser cache for some screens (including configuration) to load as expected.

> ATTENTION
> In-place migration is only supported on Windows Server 2016 systems. If your system is installed on Windows Server 2012 R2, see “Upgrading on a Windows Server R2 Server” in the APM Troubleshooting Guide before continuing.

> NOTE
> During the installation TLS 1.2 is enabled on the server.

5.2.1 Upgrading an Existing Primary Server

> NOTE
> The protocol used on the server (http or https) cannot be changed during migration. If the existing server uses http, http must be selected during the upgrade. If the existing server uses https, https must be selected during the upgrade.

> **ATTENTION**
> When migrating existing Primary and Shadow Servers, the Shadow Server Configuration Utility will look for an exact certificate match. If you are not using the original certificates for the upgrade, you must delete your existing primary and shadow server certificates on both servers before running the Shadow Server Configuration Utility.

To update a Asset Performance Management Primary Server

1. Perform an iisreset.
2. On the installation media of the new data base server, right-click **setup.exe**, and then select **Run as Administrator**.
3. If RabbitMQ is not already installed, a prompt appears. In the RabbitMQ prompt, click **Yes**.

   If RabbitMQ needs to be installed, when it completes, reopen the installation wizard as described in step 2.
4. On the Welcome page, click **Next**.
5. Read through the license agreement and click **I accept the terms in the license agreement** if you agree to the license terms and then click **Next**.
6. On the Migration Information page, click **Next**.
7. The components for the server you are installing are selected and cannot be cleared. Click **Next**.
8. On the **FeaturesandOptionsSelection**page, the features required for your components are already selected, click **Next**.
9. In the **Intuition Environment Setup,** correct any issues, and then click **Exit**. (If the Exit button does not appear with the success message, close the browser window to continue.)

   For example, if Certificate Configuration is flagged, do all of the following:
   1. Beside **Certificate Configuration**, click **Repair**.
   2. Select **Web Components** and enter the Application server name.
   3. Click **Configure**.
   4. In the **Success** dialog box, click **OK**.
   5. Click the **Home** tab and click **Repair all**.
10. On the **Notification and DAS Configuration** page, do all of the following:
    1. Under the **CMMS Alert** section, the default path where the CMMS XML files are saved is displayed. If you want to change the default path, perform the following steps:
       - Click **Browse**.
       - The **Browsefor Folder** page appears.
       - Select the path where you want save the CMMS XML files.
       - Click **OK**.
    2. Click **Next**.
11. Do all of the following:
    1. In the **Login Information for Application Services** area, in the **Account Password** box, type the password for the Asset Performance Management services user.
    2. (Optional) If you are using an existing EPHD installation, in the **EPHD Server Details** area, in the **EPHD Server Name** box, type the name of the EPHD server.

> **NOTE**
> EPHD does not support SQL Clustering.

c. In the SQL Cluster Support area, select Enable SQL Cluster to enable SQL Cluster deployment.
d. Enter the Domain User Group Name. The domain user group name should be in the doamin\UserGroupName format. For example: l4-dc\UASServiceGroup.
e. In the Service Account Permissions area, select **I accept the service account must bedenied the ability to logon to the desktop to continue**.

> **NOTE**
> EPHD does not support SQL Clustering.

f. Click **Next**.

12. Do all of the following:
    a. In the **Website selection** box, ensure the web site you are using for Asset Performance Management is selected.
    b. In the **Calc Server Name** box, type the name of the main Calculation Server.
    c. Click **Next**.
13. On the Email Dispatcher page, do all of the following:
    a. In the **Sender's Email ID**, type the email address that will appear in the From field of emails sent by the system.
    b. In the **Default Email ID**, type the email address that will receive a notification if an email delivery fails.
    c. In the **SMTP Server Name** box, type the name of the SMTP server.
    d. In the **SMTP Port Number**, type the port number used on the SMTP server.
    e. Click **Next**.
14. Click **Install** to start the installation.
15. Click **Finish**.

See Model Migration Utility Parameters for the parameters used to run the migration utility.

Model Migration Utility Parameters

Use the following syntax to run the model migration utility:

```
SentinelUpgradeConsoleUtil --address [ServerName] --[protocol] --port [PortNumber]
```

Parameter	Description
address	The name of the server.
protocol	https (when using SSL) If the server is http, do not include the protocol parameter.
port	The port number for the server.

Examples:

```
SentinelUpgradeConsoleUtil --address localhost --port 80
SentinelUpgradeConsoleUtil --address webserver1.oursite.com --https --port 8000
```

> **ATTENTION**
> If the protocol is https, localhost cannot be used as the server address.

To see a list of parameters while using the model migration utility, use --help to run the utility:

```
SentinelUpgradeConsoleUtil --help
```

5.3 Upgrading a Shadow Server

This chapter covers how to upgrade a Primary Server.

> **ATTENTION**
> After the upgrade to R530.1, users must clear their browser cache for some screens (including configuration) to load as expected.

> **ATTENTION**
> In-place migration is only supported on Windows Server 2016 systems. If your system is installed on Windows Server 2012 R2, see “Upgrading on a Windows Server R2 Server” in the APM Troubleshooting Guide before continuing.

> **NOTE**
> During the installation TLS 1.2 is enabled on the server.

5.3.1 Upgrading an Existing Shadow Server

> **NOTE**
> The protocol used on the server (http or https) cannot be changed during migration. If the existing server uses http, http must be selected during the upgrade. If the existing server uses https, https must be selected during the upgrade

> **ATTENTION**
> When migrating existing Primary and Shadow Servers, the Shadow Server Configuration Utility will look for an exact certificate match. If you are not using the original certificates for the upgrade, you must delete your existing primary and shadow server certificates on both servers before running the Shadow Server Configuration Utility.

To update a Asset Performance Management Primary Server

1. Perform an iisreset.
2. On the installation media of the new data base server, right-click **setup.exe**, and then select **Run as Administrator**.
3. If RabbitMQ is not already installed, a prompt appears. In the RabbitMQ prompt, click **Yes**.

   If RabbitMQ needs to be installed, when it completes, reopen the installation wizard as described in step 2.
4. On the Welcome page, click **Next**.
5. Read through the license agreement and click **I accept the terms in the licenseagreement**if you agree to the license terms and then click **Next**.
6. On the Migration Information page, click **Next**.
7. The components for the server you are installing are selected and cannot be cleared. Click Next.
8. On the **Features and Options Selection**page, the features required for your components are already selected, click **Next**.
9. In the **Intuition Environment Setup,** correct any issues, and then click **Exit**. (If the Exit button does not appear with the success message, close the browser window to continue.)

   For example, if Certificate Configuration is flagged, do all of the following:
   a. Beside **Certificate Configuration**, click **Repair**.
   b. Select **Web Components** and enter the Application server name.
   c. Click **Configure**.
   d. In the **Success** dialog box, click **OK**.
   e. Click the **Home** tab and click **Repair all**.
10. On the **Notification and DAS Configuration** page, do all of the following:
    a. Under the **CMMS Alert** section, the default path where the CMMS XML files are saved is displayed. If you want to change the default path, perform the following steps:
       a. Click **Browse**.
       b. The **Browsefor Folder** page appears.
       c. Select the path where you want save the CMMS XML files.
       d. Click **OK**.

    b. Click **Next**.
11. Do all of the following:
    a. In the **Login Information for Application Services** area, in the **Account Password** box, type the password for the Asset Performance Management services user.
    b. (Optional) If you are using an existing EPHD installation, in the **EPHD Server Details** area, in the **EPHD Server Name** box, type the name of the EPHD server.

       > **NOTE**
       > EPHD does not support SQL Clustering.

    c. In the SQL Cluster Support area, select Enable SQL Cluster to enable SQL Cluster deployment.
    d. Enter the Domain User Group Name. The domain user group name should be in the doamin\UserGroupName format. For example: l4-dc\UASServiceGroup.
    e. In the Service Account Permissions area, select **I accept the service account must bedenied the**

ability to logon to the desktop to continue.

> **NOTE**
> The installation cannot continue if the service account has logon permissions to the desktop. This check box must be selected for the installation to succeed. Once this change is made, the service account can no longer log on to the computer (locally or remotely) and cannot be used to run programs that interact with the desktop, such as or Internet Explorer.

f. Click **Next**.

12. Do all of the following:
    a. In the **Website selection** box, ensure the web site you are using for Asset Performance Management is selected.
    b. In the **Calc Server Name** box, type the name of the main Calculation Server.
    c. Click **Next**.
13. On the Email Dispatcher page, do all of the following:
    a. In the **Sender'sEmail ID**, type the email address that will appear in the From field of emails sent by the system.
    b. In the **Default Email ID**, type the email address that will receive a notification if an email delivery fails.
    c. In the **SMTP Server Name** box, type the name of the SMTP server.
    d. In the **SMTP Port Number**, type the port number used on the SMTP server.
    e. Click **Next**.
14. Click **Install** to start the installation.
15. Click **Finish**.
16. Do all of the following to update the calculations:
    a. Ensure the AMApplicationServer service is running.
    b. With a user that belongs to the Reliability Engineers group (see "Add required users to the Windows Reliability Engineers group" on page 154), run the **SentinelUpgradeConsoleUtil.exe** file, located in the following folder:
    `[Install Media]\Utilities\Sentinel_Upgrade_Utility\`
    See Model Migration Utility Parameters for the parameters used to run the migration utility.

Model Migration Utility Parameters

Use the following syntax to run the model migration utility:

```
SentinelUpgradeConsoleUtil --address [ServerName] --[protocol] --port [PortNumber]
```

Parameter	Description
address	The name of the server.
protocol	https (when using SSL) If the server is http, do not include the protocol parameter.
port	The port number for the server.

Examples:

```
 SentinelUpgradeConsoleUtil --address localhost --port 80
SentinelUpgradeConsoleUtil --address webserver1.oursite.com --https --port 8000
```

> **ATTENTION**
> If the protocol is https, localhost cannot be used as the server address.

To see a list of parameters while using the model migration utility, use --help to run the utility:

```
SentinelUpgradeConsoleUtil --help
```

5.4 Finalize a Primary or Shadow Server Upgrade

To complete the upgrade, so all of the following on both the Primary and Shadow Servers.

5.4.1 Confirm User Permissions and Settings

Confirm the follow are still true, and add the required permissions if not:

- Ensure the MESDBUser has Public, DBreader and DBwriter permissions to the APM AssetTaskDataModel database.
- Ensure the installation user has db owner permission to the APM AssetTaskDataModel database.
- Ensure all required users are added to the Windows Reliability Engineers group (see Add required users to the Windows Reliability Engineers group).
- Ensure Directory Browsing is enabled on the web site in IIS (see Ensure Directory Browsing is enabled for the site in IIS).

> **ATTENTION**
> After the upgrade to R540.1, users must clear their browser cache for some screens (including configuration) to load as expected.

5.4.2 Add required users to the Windows Reliability Engineers group

1. Start > Server Manager.
2. On the server tree, choose **Configuration** > **Local Users and Groups** > **Groups**.
3. Right-click the **Reliability Engineers** group, and click **Add to Group**.
4. In the **Reliability Engineers Properties** dialog box, click **Add**.
5. Enter all the names of all of the following users:

a. The services (runtime) user.

> **ATTENTION**
> If the services user is not added to the Reliability Engineers group, some features will not function as expected.

b. The install user.

c. All the users who will need to use the APM Configuration Excel Plugin and the Asset Performance Management Configuration.

6. Click **OK**. The user names appear in the group list.
7. Click **OK** to close the Reliability Engineers Properties dialog box.

> **NOTE**
> Reliability Engineers group is present in Application Server in case of split server.

5.4.3 Ensure Directory Browsing is enabled for the site in IIS

1. Start > All Apps > Internet Information Services (IIS) Manager

   Internet Information Services (IIS) Manager appears.
2. Expand Sites, and select the web site used for Asset Performance Management .
3. On Default Web Site Home page, double-click Directory Browsing.
4. In the right-hand side bar, ensure Directory Browsing is enabled.

   If it is disabled an alert reading “Directory browsing has been disabled” is visible and the Enable button is available under Actions.
5. If it is disabled, click **Enable**.
6. Perform an iisreset.

5.4.4 Update the Internal Data Source Connection Strings

If you upgraded a database and ran the Migration Utility, you need to upgrade the connections strings for the internal data sources.

> **ATTENTION**
> If the connection strings are not updated, asset metrics data can be lost on a metrics reset.

1. Open SQL Server Management Studio.
2. Expand **Database**>**Honeywell.MES.AssetTask.DataModel.AssetTaskDataModel**> **Tables**>**ExternalDataSources**.
3. Edit the table rows.
4. In the **Internal State Storage** row, in the ConnectionInformation column, replace `(local)` with the name of your Database Server. If APM is deployed on named instance on the DB machine you need to replace (local) with
5. In the **InternalHistoricalStorage**, in ConnectionInformation column, replace `(local)` with the name of your

Database Server. If APM is deployed on named instance on the DB machine you need to replace (local) with DBServer\Instance.

5.4.5 Systems Running EPHD

After installing Asset Performance Management and updating the EPHD settings, the EPHD system is left in an Initializing state. To bring the state back to active, run the following commands:

```
phdctl stop /y

phdctl stop /y dbsecurity

phdctl stop /y udbserver

phdctl start cold
```

CHAPTER

6 UPGRADING CLIENT TOOLS

6.1 Upgrading the Model Tools

This chapter describes how to upgrade Asset Performance Management R520.1 to Asset Performance Management R532.1

> **ATTENTION**
> In-place migration is only supported on Windows Server 2016/2019 systems.

6.1.1 Install all Microsoft Updates

1. Run Microsoft Updates to ensure all current updates are installed.
2. Restart the computer

6.1.2 Upgrading Model Tools

1. On the installation media, right-click **setup.exe**, and then select **Run as Administrator**. The **Welcome**page appears.
2. After you have read through the welcome information, click **Next**. The License Agreement page appears.
3. Read through the license agreement and click **I accept the terms in the license** agreement if you agree to the license terms. Click **Next**.

   The Deplment Fature Selection page appears.
4. Click **Next**.

   The **Features and Options Selection**page appears.
5. Do all of the following:
   a. Leave **Model Tool** selected.
   b. Clear **Bulk Configuration** and **PHD Utility**.
   c. Click **Next**.

   The **Application Server** Information page appears.
6. In the **Application Server details** area, do all of the following:
   a. In the **Application Server Host Name** box, type the name of the Application Server.
   b. In the **Application Server Port Number** box, type the net.tcp port number used on the Application Server.
   c. From the **Application Server Protocol Name**, select the protocol used on the Application Server (**http** or **https**).

d. In the **APM DB Server Name** box, type the name of the Database Server.
e. In the **APM Web Server Name** box, type the name of the Web Server.
f. Click Next.

The **Installation Summary** page appears.

7. Click **Install**.

   The installation begins.

8. If the **Microsoft Visual Studio Tls for Office Runtime** notification appears, click **Accept** to accept the license agreement and continue with the installation.
9. Click **Finish**.

   A message appears prompting you to restart the computer.

10. Click **Yes** to restart.

6.2 Upgrading the APM Configuration Excel Plugin

This chapter describes how to upgrade Asset Performance Management R520.1 to Asset Performance Management R540.1.

> **ATTENTION**
> In-place migration is only supported on Windows Server 2016/2019 systems.

6.2.1 Install all Microsoft Updates

1. Run Microsoft Updates to ensure all current updates are installed.
2. Restart the computer.

6.2.2 Upgrading the APMConfiguration Excel Plugin

1. On the installation media, right-click **setup.exe**, and then select **Run as Administrator**.

   The **Welcome**page appears.

2. After you have read through the welcome information, click **Next**.

   The **License Agreement** page appears.

3. Read through the license agreement and click **I accept the terms in the licenseagreement**if you agree to the license terms. Click **Next**.

   The **Deployment Feature Selection** page appears.

4. Click **Next**.

   The **Features and Options Selection**page appears.

5. Do all of the following:
   a. Leave **Bulk Configuration** selected.
   b. Clear **Model Tools** and **PHD Utility**.
   c. Click **Next**.

   The **Application Server** Information page appears.

6. In the **Application Server details** area, do all of the following:

a. In the **Application Server Host Name** box, type the name of the Application Server.

   a. In the **Application Server Port Number** box, type the net.tcp port number used on the Application Server.
   b. From the **Application Server Protocol Name**, select the protocol used on the Application Server (**http** or **https**).
   c. In the **APM DB Server Nam**e box, type the name of the Database Server.
   d. In the **APM Web Server Name** box, type the name of the Web Server.
   e. Click Next.

      The **Installation Summary** page appears.

7. Click **Install** to start the installation.

   When the installation is complete, a message appears prompting you to restart the computer.
8. Click **OK** to restart.

6.3 Upgrading the PHD Components

This chapter describes how to upgrade Asset Performance Management R520.1 to Asset Performance Management R532.1.

> **ATTENTION**
> In-place migration is only supported on Windows Server 2016/2019 systems.

6.3.1 Install all Microsoft Updates

1. Run Microsoft Updates to ensure all current updates are installed.
2. Restart the computer.

6.3.2 Upgrading the PHD Components

1. On the installation media, right-click **setup.exe**, and then select **Run as Administrator**.

   The **Welcome**page appears.
2. After you have read through the welcome information, click **Next**.

   The **License Agreement** page appears.
3. Read through the license agreement and click **I accept the terms in the license agreement** if you agree to the license terms. Click **Next**.

   The **Deployment Feature Selection** page appears.
4. Click **Next**.

   The **Features and Options Selection**page appears.
5. Do all of the following:
   a. Leave **PHD Utility**selected.
   b. Clear **Bulk Configuration** and **Model Tools**.
   c. Click **Next**.

      The **Application Server** Information page appears.
6. In the **Application Server details** area, do all of the following:

a. In the **Application Server Host Name** box, type the name of the Application Server.
b. In the **Application Server Port Number** box, type the net.tcp port number used on the Application Server.
c. From the **Application Server Protocol Name**, select the protocol used on the Application Server (**http** or **https**).
    a. In the **APM DB Server Nam**e box, type the name of the Database Server.
    b. In the **APM Web Server Name** box, type the name of the Web Server.
    c. Click **Next**.

       The **Installation Summary** page appears.

7. Click **Install**.

   The installation begins.
8. Click **Finish**.

   A message appears prompting you to restart the computer.
9. Click **Yes** to restart.

CHAPTER

7 MIGRATING TO A NEW SYSTEM

7.1 Migrating Split Servers to New Systems

This chapter explains how to move existing Asset Performance Management servers to a new system with Asset Performance Management R532.1 installed.

> **NOTE**
> Migrating to a new system requires a fresh installation of Asset Performance Management on the new server. For detailed instructions for installing Asset Performance Management , see the Asset Performance Management Installation Guide.

7.1.1 Confirm User and Services Settings

On all new servers, ensure the following settings are in place:

- The installation user is a Local Administrator and has SQL Server (SysAdmin) privileges.
- Net 4.7.2 is installed on the system.
- The Windows Update service is disabled. (**Run** > **Services**, right-click **Windows Update** and then select **Disable**.)
- Ensure you have a Services User that does not require logon permissions to the Asset Performance Management computer.

> **ATTENTION**
> Do not use the installation (current) user as the Services User. If the selected Services User has logon permissions, they will be removed as a part of the Asset Performance Management installation. Once the installation is complete, the Services User will no longer be able to log on to the computer locally or remotely.

- Ensure the services (runtime) is added to the MESDBUsers Windows Group.
- Ensure all required users are added to the Windows Reliability Engineers group (Add required users to the Windows Reliability Engineers group”).
- Ensure the services account is given the following permissions to the [InstallDrive]:\ProgramData\Microsoft\Crypto\RSA\MachineKeys folder on the Primary Server:
  - Read & Execute
  - List Folder contents
  - Read

7.1.2 Confirm SQL Server Permissions and Settings

On a new database server, ensure the following settings are in place:

- Ensure the installation user has db_owner permission to the APM AssetTaskDataModel database.
- Ensure the installation user has public, db_datareader and db_datawriter on all Honeywell MES databases.
- Ensure the MESDBUser has public, db_datareader and db_datawriter permissions to the APM AssetTaskDataModel database.
- Delete the services user account in SQL.
- The migration adds the historical data clean-up SQL job (see Configure Historized Data Clean-up in the Asset Performance Management Configuration Guide for details). Ensure the SQL Server Agent service is running either during or after the migration for this change to be completed.

> ATTENTION
> Ensure SQL replication is disabled before beginning the migration.

7.1.3 Install all Microsoft Updates

On all new servers, do all of the following:

1. Run Microsoft Updates to ensure all current updates are installed.
2. Restart the server.

7.1.4 Back-up Existing Files

On the old servers, create a back-up copy of the database and any custom files before beginning the upgrade.

> NOTE
> For instructions on backing-up a database through SQL Server Management Studio, see the SQL Server Management Studio user documentation.

1. Take back up from

   [InstallDrive]:\Program File (x86)\Honeywell\MES\AssetManager\AppServer\Bin\Honeywell.AM.Service.AMHostingService.exe.config

   [InstallDrive]:\Program Files (x86)\Honeywell\MES\AssetManager\CalcHost\Honeywell.UAS.CalculationWorker.exe.config
2. Browse to **[InstallDrive]:\Honeywell\MES\Core\WWWMESCore\AMInformationFolders**, copy the folders: **Images**, **Icons**, and **Documents**.
3. Paste the copy of the folders to any folder or external drive on your computer.

7.1.5 Migrating All Asset Performance Management Split Servers

You can move one server in the split server topology to new hardware, or move all servers at once. If migrating multiple servers, move the systems in the following order:

1. Database Server: See Migrating Asset Performance Management to a New Database Server.
2. Application Server: See Migrating Asset Performance Management to a New Application Server.
3. Web Server: See Migrating Asset Performance Management to a New Web Server.
4. Calculation Server(s): See Migrating Asset Performance Management to a New Calculation Server.

7.1.6 Migrating Asset Performance Management to a New Database Server

1. On the existing Database Server back-up your existing databases.

   For a complete list of databases, see Backing-up a Database.

2. On the installation media of the new data base server, right-click **setup.exe**, and then select **Run as Administrator**.
3. Follow the steps in the installation wizard to install R532.1. For detailed steps, see in "Installing a Database Server" in the Asset Performance Management Installation Guide.
4. Copy and restore the old database onto the new Database Server.
5. Confirm the following are still true, and add the required permissions if not:
   a. Ensure the installation user has db owner permission to the APM AssetTaskDataModel database.
   b. Ensure the MESDBUser has Public, DBreader and DBwriter permissions to the APM AssetTaskDataModel database.

> **NOTE**
> To check how the db migration utility expects command navigate to the Upgrade folder, and run the UAS_DatabaseMigrationUtility.bat file, using the following command UAS_DatabaseMigrationUtility.bat help.

6. On the installation media, on the new Database Server, navigate to the Upgrade folder, and run the UAS_DatabaseMigrationUtility.bat file, using the following command to migrate a UAS R520, UAS R530, UAS R531, UAS R532.1 database to APM 540.1 (replace SQLServerName\InstanceName with the name of new database instance):UAS_DatabaseMigrationUtility.bat SQLServerName\InstanceName Domain\ GroupName Upgrade 520

   For the Domain\GroupName parameter:

   a. If it is a Single Server Deployment, the input for the parameter is ServerName\ MESDBUser.

      For example: UAS_DatabaseMigrationUtility.bat UASAuto11 UASAuto11\MESDBUser Upgrade

      For named instance:UAS_DatabaseMigrationUtility.bat UASAuto11\Sentinel1 UASAuto11\MESDBUser Upgrade

   b. If it is a Split Server Deployment, the input for the parameter is DBServerName\ MESDBUser.

      For example: UAS_DatabaseMigrationUtility.bat UAS_2019_DB UAS_2019_DB\ MESDBUser Upgrade

      For the named instance: UAS_DatabaseMigrationUtility.bat UAS_2019_DB\Sentinel1 UAS_2019_DB\MESDBUser Upgrade

   c. If the SQL Cluster option was selected during installation the input for the parameter is domain\UserGroupName.

      For example: UAS_DatabaseMigrationUtility.bat UASAuto11 L4-DC\UASServiceGroup Upgrade 520

      For the named instance: UAS_DatabaseMigrationUtility.bat UASAuto11\Sentinel1 L4-DC\UASServiceGroup Upgrade

7. After migrating to APM R540.1, ensure that you run the CAM "SystemcatalogSeedUtility.exe" utility located on the **Application Server**. Typically Path-C:\Program Files (x86)\Honeywell\MES\Tools\CAM\SystemcatalogSeedUtility\The above step must be run before running the next command line step
8. After running the Migration Utility, do all of the following to upgrade the connections strings for the internal datasources on the Database Server:

> **ATTENTION**
> If the connection strings are not updated, asset metrics data can be lost on a metrics reset.

   a. Open SQL Server Management Studio.
   b. Expand **Databases** >**Honeywell.MES.AssetTask.DataModel.AssetTaskDataModel**>**Tables**> **ExternalDataSources**.
   c. Edit the table rows.
   d. In the **Internal State Storage** row, in the ConnectionInformation column, verify SQL Connection String data source should point to your Migrated DBServer\Instance.
   e. In the **Internal Historical Storage**, in ConnectionInformation column, verify SQL Connection String data source should point to your Migrated DBServer\Instance.

9. On the installation media, navigate to the Upgrade folder, and run the UAS_ DatabaseMigrationUtility.bat file, using the following command: UAS_DatabaseMigrationUtility.bat SQLServerName\InstanceName Domain\ GroupName PostMigration

   For the Domain\GroupName parameter:

   a. If it is a Single Server Deployment, the input for the parameter is ServerName\ MESDBUser.

      For example: UAS_DatabaseMigrationUtility.bat UASAuto11 UASAuto11\MESDBUser PostMigration

      For named instance:UAS_DatabaseMigrationUtility.bat UASAuto11\Sentinel1 UASAuto11\MESDBUser PostMigration

   b. If it is a Split Server Deployment, the input for the parameter is DBServerName\ MESDBUser.

      For example: UAS_DatabaseMigrationUtility.bat UAS_2019_DB UAS_2019_DB\ MESDBUser PostMigration

      For the named instance: UAS_DatabaseMigrationUtility.bat UAS_2019_DB\Sentinel1 UAS_2019_ DB\MESDBUser PostMigration

   c. If the SQL Cluster option was selected during installation the input for the parameter is domain\UserGroupName.

      For example: UAS_DatabaseMigrationUtility.bat UASAuto11 L4-DC\UASServiceGroup PostMigration

      For the named instance: UAS_DatabaseMigrationUtility.bat UASAuto11\Sentinel1 L4-DC\UASServiceGroup PostMigration

10. If Database Migration is being done after software install. Run the following PowerShell script as an administrator on the Application server, Web server and calculation server as follows:

    a. On the installation media navigate to Utilities\EncryptData.
    b. Use **Run as Administrator** to open PowerShell run the following command.

       .\EncryptData.ps1

    c. Perform iisreset.

11. Perform the post install steps Certificate Exchange, Post installation for RabbitMQ, Configuring SSL certificate, refer to installation guide. Import, Export and Influxdb services should be running status.

Re-register the Calculation Servers

As Asset Performance Management deployments can have multiple calculation servers, entires for the Calculation Servers in the restored database are not over-written. To ensure the list of Calculation Servers is correct for your current deployment, you must remove the existing entires and redistribute the calculations.

> NOTE
> If you are also moving your Calculation Servers to new systems, complete the calculation server changes before distributing the calculations.

To update the list of calculations servers after database change

1. On the Database Server, on the installation media, navigate to the following folder:
   [installationedia]:\Upgrade\
2. Open the UAS_ResetScaleOutServers.ps1 file with a text editor.
3. Edit the **ServerInstanceName** detail with the Servername and instance.
4. Save and close the file.
5. Run the follow PowerShell script as an administrator:
   UAS_ResetScaleOutServers.ps1
6. Once all the calculations servers are upgraded, moved, or installed, do all of the following to distribute the calculations:
   a. Open **Asset Performance Management** , and on the Contents tree choose **System Console**. The **System Console** page appears.
   b. In the **Distribute Calculations** section, click **Distribute.**
      A status message is displayed as the calculations are distributed.

Systems Running EPHD

After installing Asset Performance Management and updating the EPHD settings, the EPHD system is left in an Initializing state. To bring the state back to active, run the following commands:

```
phdctl stop /y

phdctl stop /y dbsecurity

phdctl stop /y udbserver

phdctl start cold
```

7.1.7 Restoring EPHD archive data

To restore EPHD machine data,

1. Stop all the Uniformance PHD services.
2. Go to services, right click Uniformance PHD Server and slect properties.
3. Click General and navigate to the path from path to executable.
4. Click Archive and rename all the file name from CHAR00001 to CHAR00002.
5. Go to :\Program Files (x86)\Honeywell\Uniformance\PHDServer\Archive and copy all the files from the location.
6. Navigate back to Path to executable location of the Uniformance PHD Server properties and paste.
7. Start all the Uniformance PHD Services back.

7.1.8 Migrating Asset Performance Management to a New Application Server

1. On the installation media of the new data base server, right-click setup.exe, and the select Run as Administrator.
2. Follow the steps in the installation wizard to install R532.1. For detailed steps, see in “Installing an Application Server” in the Asset Performance Management Installation Guide.
3. Run the SystemCatalogSeedUtility.exe file located in the following folder:

```
[install drive]:\Program Files
(x86)\Honeywell\MES\Tools\CAM\SystemCatalogSeedUtility\
```

7.1.9 Migrating Asset Performance Management to a New Web Server

1. On the installation media of the new data base server, right-click **setup.exe**, and then select **Run as Administrator**.
2. Follow the steps in the installation wizard to install R532.1. For detailed steps, see in “Installing a Web Server” in the Asset Performance Management Installation Guide.
3. Do all of the following to update the calculations:
   a. Ensure the AMApplicationServer service is running and Asset Performance Management is licensed.
   b. With a user that belongs to the Reliability Engineers group (see “Add required users to the Windows Reliability Engineers group” on page 154), run the **SentinelUpgradeConsoleUtil.exe** file, located in the following folder:

   ```
   [Install Media]\Utilities\Sentinel_Upgrade_Utility\
   ```

   See Model Migration Utility Parameters for the parameters used to run the migration utility

Model Migration Utility Parameters

Use the following syntax to run the model migration utility:

```
SentinelUpgradeConsoleUtil --address [ServerName] --[protocol] --port [PortNumber]
```

Parameter	Description
address	The name of the server.
protocol	https (when using SSL) If the server is http, do not include the protocol parameter.
port	The port number for the server.

Examples:

```
SentinelUpgradeConsoleUtil --address localhost --port 80
SentinelUpgradeConsoleUtil --address webserver1.oursite.com --https --port 8000
```

> **ATTENTION**
> If the protocol is https, localhost cannot be used as the server address.

To see a list of parameters while using the model migration utility, use --help to run the utility:

```
SentinelUpgradeConsoleUtil --help
```

Ensure Directory Browsing is enabled for the site in IIS

1. **Start** > **All Apps** > **Internet Information Services (IIS) Manager** Internet Information Services (IIS) Manager appears.
2. Expand Sites, and select the web site used for Asset Performance Management .
3. On Default Web Site Home page, double-click Directory Browsing.
4. In the right-hand side bar, ensure Directory Browsing is enabled.

   If it is disabled an alert reading “Directory browsing has been disabled” is visible and the Enable button is available under Actions.
5. If it is disabled, click **Enable**.
6. Perform an iisreset.

7.1.10 Migrating Asset Performance Management to a New Calculation Server

1. On the installation media of the new data base server, right-click **setup.exe**, and then select **Run as Administrator**.
2. Follow the steps in the installation wizard to install R532.1. For detailed steps, see in “Installing a Calculation Server” in the Asset Performance Management Installation Guide.

Add required users to the Windows Reliability Engineers group

1. **Start**> **Server Manager**.
2. On the server tree, choose **Configuration** > **Local Users and Groups** > **Groups**.
3. Right-click the **Reliability Engineers** group, and click **Add to Group**.
4. In the **Reliability Engineers Properties** dialog box, click **Add**.
5. Enter all the names of all of the following users:
   a. The services (runtime) user.

      > **ATTENTION**
      > If the services user is not added to the Reliability Engineers group, some features will not function as expected.

   b. The install user.
   c. All the users who will need to use the APM Configuration Excel Plugin and the Asset Performance Management Configuration.
6. Click **OK**.

   The user names appear in the group list.
7. Click **OK** to close the Reliability Engineers Properties dialog box.

7.2 Migrating a Primary Server to a New System

This chapter describes how to upgrade and move an existing Asset Performance Management primary server to a new system with Asset Performance Management R532.1 installed.

> NOTE
> Migrating to a new system requires a fresh installation of Asset Performance Management on the new server. For detailed instructions for installing Asset Performance Management , see the *Asset Performance Management Installation Guide*.

7.2.1 Migrating Asset Performance Management to a New Primary Server

1. Back-up your existing databases on the old primary server.

   For a complete list of databases, see Backing-up a Database .
2. On the installation media of the new data base server, right-click **setup.exe**, and then select **Run as Administrator**.
3. Follow the steps in the installation wizard to install R532.1. For detailed steps, see in "Installing a Primary Server" in the Asset Performance Management Installation Guide.
4. Restore the old database onto the new server.
5. On the installation media, navigate to the Upgrade folder, and run the UAS_ DatabaseMigrationUtility.bat file, using the following command to migrate a UAS R520 or UAS R530 or UAS R531 or UAS 532.1 database to APM R540.1 (replace SQLServerName\InstanceName with the name of new database instance): UAS_ DatabaseMigrationUtility.bat SQLServerName\InstanceName Domain\ GroupName Upgrade
6. After running the Database Migration Utility, do all of the following to upgrade the connections strings for the internal data source:

   > ATTENTION
   > If the connection strings are not updated, asset metrics data can be lost on a metrics reset.

   a. Open **SQL Server Management Studio**.
   b. Expand **Databases** > **Honeywell.MES.AssetTask.DataModel.AssetTaskDataModel** >**Tables** > **ExternalDataSources**.
   c. Edit the table rows.
   d. In the **Internal State Storage** row, in the ConnectionInformation column, replace (local) with the name of your Database Server.
7. Do all of the following to update the calculations:
   a. Ensure the AMApplicationServer service is running.
   b. With a user that belongs to the Reliability Engineers group (see Add required users to the Windows Reliability Engineers group), run the **SentinelUpgradeConsoleUtil.exe** file, located in the following folder:

   ```
   [Install Media]\Utilities\Sentinel_Upgrade_Utility\
   ```

   See Model Migration Utility Parameters for the parameters used to run the migration utility.

Model Migration Utility Parameters

> NOTE
> Run the database migration utility only if you are migrating from R510 to R511.

Use the following syntax to run the model migration utility:

```
SentinelUpgradeConsoleUtil --address [ServerName] --[protocol] --port [PortNumber]
```

Parameter	Description
address	The name of the server.
protocol	https (when using SSL) If the server is http, do not include the protocol parameter.
port	The port number for the server.

Examples:

```
SentinelUpgradeConsoleUtil --address localhost --port 80 SentinelUpgradeConsoleUtil --address webserver1.oursite.com --https --port 8000
```

> **ATTENTION**
> If the protocol is https, localhost cannot be used as the server address.

To see a list of parameters while using the model migration utility, use --help to run the utility:

```
SentinelUpgradeConsoleUtil --help
```

On the installation media, navigate to the Upgrade folder, and run the `UAS_ DatabaseMigrationUtility.bat file`, using the following command:

```
UAS_DatabaseMigrationUtility.bat SQLServerName\InstanceName PostMigration
```

Confirm the follow are still true, and add the required permissions if not:

- Ensure the MESDBUser has Public, DBreader and DBwriter permissions to the APM AssetTaskDataModel database.
- Ensure the installation user has db owner permission to the APM AssetTaskDataModel database.

7.3 Migrating a Shadow Server to a New System

This chapter explains how to upgrade and move an existing Asset Performance Management shadow server to a new system with Asset Performance Management R532.1 installed.

> **NOTE**
> Migrating to a new system requires a fresh installation of Asset Performance Management on the new server. For detailed instructions for installing Asset Performance Management , see the *Asset Performance Management Installation Guide*.

7.3.1 Migrating a Single Server to a New System

This chapter describes how to upgrade and move an existing Asset Performance Management single server to a new system with Asset Performance Management R532.1 installed.

> **NOTE**
> Migrating to a new system requires a fresh installation of Asset Performance Management on the new server. For detailed instructions for installing Asset Performance Management , see the *Asset Performance Management Installation Guide*.

7.3.2 Migrating Asset Performance Management to a New Single Server

1. Back-up your existing databases on the old single server.

   For a complete list of databases, see Backing-up a Database .
2. On the installation media of the new data base server, right-click **setup.exe**, and then select **Run as Administrator**.
3. Follow the steps in the installation wizard to install R532.1. For detailed steps, see in "Installing a Single Server" in the Asset Performance Management Installation Guide.
4. Restore the old database onto the new server.
5. On the installation media, navigate to the Upgrade folder, and run the UAS_ DatabaseMigrationUtility.bat file, using the following command to migrate a R520 or UAS R530 or UAS R531 or UAS 532.1 database to APM 540.1 (replace SQLServerName\InstanceName with the name of new database instance): UAS_ DatabaseMigrationUtility.bat SQLServerName\InstanceName Domain\ GroupName Upgrade 520
6. After running the Database Migration Utility, do all of the following to upgrade the connections strings for the internal data sources:

   > **ATTENTION**
   > If the connection strings are not updated, asset metrics data can be lost on a metrics reset.

   a. Open **SQL Server Management Studio**.

   b. Expand **Databases** > **Honeywell.MES.AssetTask.DataModel.AssetTaskDataModel** >**Tables** > **ExternalDataSources**.

   c. Edit the table rows.

   d. In the **Internal State Storage** row, in the ConnectionInformation column, replace `(local)` with the name of your Database Server.

   e. In the **Internal Historical Storage**, in ConnectionInformation column, replace `(local)` with the name of your Database Server.
7. Run the **SystemCatalogSeedUtility.exe** file located in the following folder:

   ```
   [install drive]:\Program Files
   (x86)\Honeywell\MES\Tools\CAM\SystemCatalogSeedUtility\
   ```
8. Do all of the following to update the calculations:

   a. Ensure the AMApplicationServer service is running and Asset Performance Management is licensed.

   b. With a user that belongs to the Reliability Engineers group (see "Add required users to the Windows Reliability Engineers group"), run the **SentinelUpgradeConsoleUtil.exe** file, located in the following folder:

   ```
   [Install Media]\Utilities\Sentinel_Upgrade_Utility\
   ```

   See Model Migration Utility Parameters for the parameters used to run the migration utility.

Model Migration Utility Parameters

Use the following syntax to run the model migration utility:

```
SentinelUpgradeConsoleUtil --address [ServerName] --[protocol] --port [PortNumber]
```

Parameters	Description
address	The name of the server.
protocol	https (when using SSL)

Parameters	Description
	If the server is http, do not include the protocol parameter.
port	The port number for the server.

Examples:

```
SentinelUpgradeConsoleUtil --address localhost --port 80 SentinelUpgradeConsoleUtil -
-address webserver1.oursite.com --https --port 8000
```

> **ATTENTION**
> If the protocol is https, localhost cannot be used as the server address.

To see a list of parameters while using the model migration utility, use --help to run the utility:

```
SentinelUpgradeConsoleUtil --help
```

On the installation media, navigate to the Upgrade folder, and run the `UAS_ DatabaseMigrationUtility.bat` file, using the following command:

```
UAS_DatabaseMigrationUtility.bat SQLServerName\InstanceName PostMigration
```

Confirm the follow are still true, and add the required permissions if not:

- Ensure the MESDBUser has Public, DBreader and DBwriter permissions to the APM AssetTaskDataModel database.
- Ensure the installation user has db owner permission to the APM AssetTaskDataModel database.

7.4 InfluxDB Migration

> **ATTENTION**
> The following steps are not required while migrating from R532.3 and above versions.

To take backup of InfluxDB follow the below steps:

1. Perform the following steps in the DataBase Server:
    a. Run the command prompt as administrator.
    b. Navigate to InfluxDB install location.
       <Install Location>:\Program Files\influxData\influxdb
    c. Run **influxd backup -portable <path-to-backup-folder>** in command prompt.
       <path-to-backup-folder> is the path to InfluxDB folder location.

       > **TIP**
       > Ensure that all the files are generated in the above path.

2. Perform the following steps in the Application Server:

a. Navigate to the following influx migration utility folder:

   <Install Location>\Program Files (x86)\Honeywell\ForgeApps\APM\ConfigurationService\InfluxDBMigUtil\

b. Follow the below steps:

   i. Open the command prompt as administrator.

   ii. Enter **AssetAnalytics.Influxdb.DataMigration.exe** in command prompt and press **Enter**.

      The migration starts.

   iii. Once the migration is completed, log is generated in the below path:

      c://Influxdb-data-migration/logs

      The log path can be changed in appsettings.json

c. Now you can validate the 'VariableResults' and 'ModelResults' measurements data with the updated details.

CHAPTER

8 RESTORING A DATABASE

8.1 Backing-up a Database

This chapter describes the various databases that you must back up before restoring Asset Performance Management on a new system.

For more information on how to do a periodic backup and schedule a job for backups, you can refer to the SQL backup procedures provided in the SQL Server help.

Database	Path of the Database
Honey- well.MES.AssetTask.DataModel. AssetTaskDataModel	<Program Files>\Microsoft SQL Serv-er\<MSSQLSERVER Path>\MSSQL\DATA

> NOTE
> For Intuition and PHD migration, contact Honeywell support team.

8.2 Restoring a Database

This chapter explains the procedure to restore a database from an earlier version into a new system. You must have Asset Performance Management R511.1 or later installed on the system to perform a database restore, however, the database restore is supported for database versions from Asset Manager R410.2 or Asset Performance Management R500.x to R540.1.

> ATTENTION
> The collation on the new Database Server and the database to be restored must match.

To restore the Asset Performance Management database

1. On the new Database Server, do all of the following:
   a. From the **Services** window, stop **AMApplictionServer** and **AssetManagerIntegrationService** service.

   > ATTENTION
   > Ensure that the Asset Performance Management User Interface is closed.

   b. In SQL Server Management Studio, restore all the databases from the backup.

   > ATTENTION

> Under Restore Job Properties > Settings > Microsoft SQL, ensure that you select the Overwrite the existing database option when you are restoring the PHDAPP and PHDCFG databases.
>
> If you encounter any failure in the database restore activity, you must restart the SQL services and try again.

2. On the installation media, on the new Database Server, navigate to the Upgrade folder, and run the UAS_ DatabaseMigrationUtility.bat file, using the following command to migrate a UAS R510, UAS R511 or UAS R520 or UAS R530 or UAS 531.1 database to APM R540.1 (replace SQLServerName\InstanceName with the name of new database instance): UAS_DatabaseMigrationUtility.bat SQLServerName\InstanceName Domain\ GroupName Upgrade 520

   When the installation is complete, see Update the Internal Datasource ConnectionStrings.
3. On the Application Server, run the **SystemCatalogSeedUtility.exe** file located in the following folder:

   ```
   [install drive]:\Program Files
   (x86)\Honeywell\MES\Tools\CAM\SystemCatalogSeedUtility\
   ```
4. On the new Database Server, on the installation media, navigate to the Upgrade folder, and run the UAS_ DatabaseMigrationUtility.bat file, using the following commad:

   ```
   UAS_DatabaseMigrationUtility.bat SQLServerName\InstanceName PostMigration
   ```

Re-register the Calculation Servers

As Asset Performance Management deployments can have multiple calculation servers, entires for the Calculation Servers in the restored database are not over-written. To ensure the list of Calculation Servers is correct for your current deployment, you must remove the existing entires and redistribute the calculations.

> NOTE
> If you are also moving your Calculation Servers to new systems, complete the calculation server changes before distributing the calculations.

To update the list of calculations servers after database change

1. On the Database Server, on the installation media, navigate to the following folder:

   ```
   [installationedia]:\Upgrade\
   ```
2. Open the UAS_ResetScaleOutServers.ps1 file with a text editor.
3. Edit the **ServerInstanceName** detail with the Servername and instance.
4. Save and close the file.
5. Run the follow PowerShell script as an administrator:

   ```
   UAS_ResetScaleOutServers.ps1
   ```
6. Once all the calculations servers are upgraded, moved, or installed, do all of the following to distribute the calculations:
   a. Open **Asset Performance Management**, and on the Contents tree choose **System Console**. The **System Console** page appears.
   b. In the **Distribute Calculations** section, click **Distribute.**

      A status message is displayed as the calculations are distributed.

Update the Internal Datasource Connection Strings

If you restored a database and ran the Migration Utility, you need to upgrade the connections strings for the internal data sources on the Database Server.

> **NOTE**
> If the connection strings are not updated, asset metrics data can be lost on a metrics reset.

To update the internal datasource connection strings

1. Open **SQL Server Management Studio**.
2. Expand **Databases** > **Honeywell.MES.AssetTask.DataModel.AssetTaskDataModel** > **Tables** > **ExternalDataSources**.
3. Edit the table rows.
4. In the **Internal State Storage** row, in the ConnectionInformation column, replace `(local` with the name of your Database Server. If APM is deployed on named instance on the DB machine you need to replace (local) with DBServer\Instance.
5. In the **Internal Historical Storage**, in ConnectionInformation column, replace `(local)` with the name of your Database Server. If APM is deployed on named instance on the DB machine you need to replace (local) with DBServer\Instance.

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